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3-1

能在 1min 测出并显示出恒定水温的 98%，这里的 “恒定水温” 属于单位阶跃输入，因此，该题分析的是一阶惯性系统的单位阶跃响应，对于一阶惯性系统，其阶跃响应为 $y(t)=1-e^{-t/\tau}$ 当$y(t)=0.98$即$0.98=1-e^{-t/\tau}$ 解得$\tau=\dfrac{1}{3.912}\approx 0.256\ \text{min}$

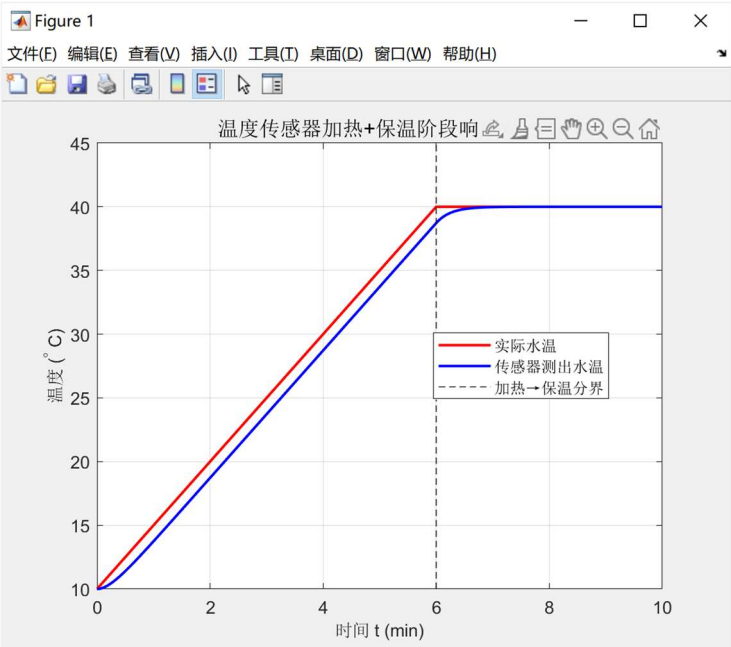
实际温度$0\le t\le 6\ \text{min}$时$T(t)=10+5t$；$t>6\ \text{min}$时$T(t)=40^{\circ}\text{C}$

传感器测量值：一阶系统对斜坡输入$r(t)=at+b$的响应为$y(t)=at+b-a\tau(1-e^{-t/\tau})$

这里$a=5$，$b=10$，$\tau=0.256\ \text{min}$

$$\therefore T_s(t)=\begin{cases}10+5t-1.28(1-e^{-3.912t})(0\le t\le 6\ \text{min})\\ 40-30e^{-t/0.256}\left(t>6\text{min}\right)\end{cases}$$

测量误差：$e(t)=\begin{cases}1.28(1-e^{-3.912t})(0\le t\le 6\ \text{min})\\ 30e^{-t/0.256}\left(t>6\text{min}\right)\end{cases}$



3-2已知$x_o^{斜坡}(t)=t-0.1[1+e^{-0.5t}(sin3.11t-cos3.11t)](t\ge 0)$

单位阶跃响应为

$$x_u(t)=\frac{dt}{d}x_o^{斜坡}(t)=\frac{dt}{d}(t)-\frac{dt}{d}(0.1)-0.1\times\frac{dt}{d}[e-0.5t(\sin 3.11t-\cos 3.11t)]$$

$$=1-0.1e^{-0.5t}(3.61cos3.11t+2.61sin3.11t)$$

单位脉冲响应为

$$x_\delta(t)=\frac{dt}{d}x_u(t)=-0.1\cdot\frac{dt}{d}[e^{-0.5t}(3.61cos3.11t+2.61sin3.11t)]$$
$$=-0.1\cdot e^{-0.5t}(6.312\cos 3.11t-12.532\sin 3.11t)$$

3-5

（1）系统闭环传递函数 $\Phi_B(s)=\dfrac{9}{s^2+s+9}$

$$s^2+2\xi\omega_n s+\omega_n^2$$

二阶系统标准形式为 $\omega_n^2=9\Rightarrow\omega_n=3\text{ rad/s}$

$$2\xi\omega_n=1\Rightarrow\xi=\frac{1}{2\omega_n}=\frac{1}{6}\approx 0.167$$

（2）最大超调量 $M_p=e^{-\pi\xi/\sqrt{1-\xi^2}}=e^{-\pi\times 0.167/\sqrt{1-0.167^2}}\approx 60.5\%$

峰值时间 $t_p=\dfrac{\pi}{\omega_n\sqrt{1-\xi^2}}=\dfrac{\pi}{3\sqrt{1-(1/6)^2}}\approx 1.06\text{ s}$

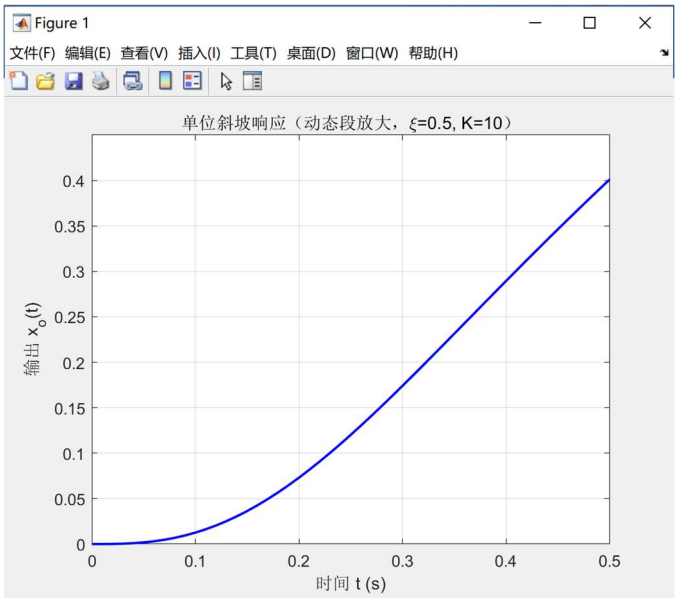
调整时间 $t_s=\dfrac{3}{\xi\omega_n}=\dfrac{3}{0.167\times 3}\approx 6\text{ s}(\Delta=5\%)$

振荡次数 $N=\dfrac{\sqrt{1-\xi^2}}{2\pi\xi}ln\dfrac{1}{\Delta\sqrt{1-\delta^2})}\approx 2.8$

3-7

$$\Phi_B(s)=\frac{K}{0.1s^2+s+K}=\frac{10K}{s^2+10s+10K}$$

$\because 2\xi\omega_n=10,\xi=0.5\therefore\omega_n=10\text{ rad/s}\because\omega_n^2=10K\therefore K=10$



3-8 闭环传递函数$\Phi(s)=\dfrac{K}{s^2+KK_f s+K}$

$\because s^2+2\xi\omega_n s+\omega_n^2 \Rightarrow 2\xi\omega_n = KK_f, \omega_n^2 = K$

$M_p = 25\% = e^{-\pi\xi/\sqrt{1-\xi^2}} \Rightarrow \xi \approx 0.403$

$t_p = 2\,\mathrm{s} = \dfrac{\pi}{\omega_n\sqrt{1-\xi^2}} \Rightarrow \omega_n \approx 1.64 rad/s$

$\Rightarrow K \approx 2.69 \quad K_f \approx 0.49$

3-9

$\because G_k(s)=\dfrac{K}{s(s+1)(s+5)}$型次为1

又$\because$是单位斜坡输入 $e_{ss}=\dfrac{1}{K_v}$

已知$e_{ss}=0.01 \quad K_v = lim_{s\to 0} s\cdot\dfrac{K}{s(s+1)(s+5)} = \dfrac{K}{5}$

$\dfrac{5}{K}=0.01 \Rightarrow K=\dfrac{5}{0.01}=500$

3-11

（1）

误差$E(s) = X_i(s) - X_o(s)$，令$X_i(s)=0$，则$E(s) = -X_o(s)$

$N(s)=2u(\mathrm{t})\therefore$拉氏变换为$\dfrac{2}{s}$

$$\frac{X_o(s)}{N(s)}=\frac{\dfrac{4}{(s+4)}}{1+10\cdot\dfrac{4}{s(s+4)}}=\frac{4}{s^2+4s+40}$$

$$e_{ssn}=lim_{s\to 0} s\cdot E(s)=lim_{s\to 0} s\cdot(-\frac{2\cdot 4}{s(s^2+4s+40)})=0$$

（2）输入的拉氏变换$X_i(s)=\dfrac{1}{s^2}$

输入到误差的传递函数$\dfrac{E(s)}{X_i(s)}=\dfrac{1}{1+G(s)}=\dfrac{s^2+4s}{s^2+4s+40}$

误差的拉氏变换$E(s)=\dfrac{s^2+4s}{s^2+4s+40}\cdot\dfrac{1}{s^2}$

终值定理求稳态误差$e_{ssi}=lim_{s\to 0}s\cdot E(s)=lim_{s\to 0}\frac{s\ \ +4}{s^2+4s+40}=0.1$

总稳态误差为两者之和：$e_{\mathrm{ss}}=e_{\mathrm{ssi}}+e_{\mathrm{ssn}}=0.1+0=0.1$

3-12

（1）输入Xi(s)作用下的稳态误差

闭环误差传递函数$\Phi_{ei}(s)=\frac{1}{1+G_k(s)}=\frac{4s+1}{4s+5}$

稳态误差$e_{ssi}=\lim\limits_{s\to 0}s\cdot X_i(s)\cdot\Phi_{ei}(s)=\lim\limits_{s\to 0}s\cdot\frac{1}{s}\cdot\frac{4s+1}{4s+5}=0.2$

（2）扰动N(s)作用下的稳态误差

扰动误差传递函数$\Phi_{en}(s)=-\frac{\frac{1}{3s+1}}{1+\frac{4}{4s+1}}$

稳态误差$e_{ssn}=\lim\limits_{s\to 0}s\cdot N(s)\cdot\Phi_{en}(s)=\lim\limits_{s\to 0}s\cdot\frac{1}{s}\cdot\left(-\frac{\frac{1}{3s+1}}{1+\frac{4}{4s+1}}\right)==-0.2$